

LESSON 4.7

FINDING RADIUS AND DIAMETER



LESSON INTRODUCTION

MA.7.GR.1.3 Explore the proportional relationship between circumferences and diameters of circles. Apply a formula for the circumference of a circle to solve mathematical and real-world problems.

LEARNING TARGETS

- I can find the radius and diameter of circles, given the circumference.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

N/A

ESSENTIAL QUESTIONS

- How can you use the circumference to find the radius?
- How can you use the circumference to find the diameter?

LESSON NARRATIVE

In the previous lesson, students used dimensions of radius or diameter to find the circumference of a circle. Students extend the flexibility of their working definitions of the circumference formula to transition from finding circumference given radius or diameter to finding radius or diameter given circumference. All questions are set in real-world contexts.

COMPONENT SUMMARY

4.7.1 WARM-UP (~5 MIN.)

Convert centimeters and millimeters to inches for comparison

4.7.3 GUIDED PRACTICE (10 MIN.)

Practice using circumference to find the radius and diameter

4.7.5 WRAP-UP (~10 MIN.)

Demonstrate ability to use the circumference to find the diameter and radius

4.7.2 EXPLORATION (15 MIN.)

Explore how to use the circumference to find the radius and diameter

4.7.4 YOUR TURN! (10 MIN.)

Additional practice using the circumference to find the diameter and radius

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Circumference
- Radius
- Diameter

MATERIALS

- (Optional) Chart paper & markers
- Colored writing utensils
- Calculators

ADDITIONAL PREPARATION

- Consider modifying the people, places, and things in the real-world problems in 4.7.3 Guided Practice and 4.7.4 Your Turn! to connect them to students' interests and everyday lives.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may divide $\frac{\pi}{C}$ instead of $\frac{C}{\pi}$ to calculate the diameter.
- Students may not divide by two when using the circumference to find the radius.

THINGS TO CONSIDER

- Consider creating an anchor chart with the formulas for circumference, radius, and diameter. Include visuals and examples for students to reference.
- Establish with your students what form of pi you would prefer that they use for the lesson. Also, establish the place value to which you expect that they round their answer.

LITERACY AND LANGUAGE ROUTINES

Compare and Connect

In 4.7.2 Exploration, students compare and contrast the relationships between the radius, diameter, and circumference of circles. Prompt a variety of students to share their thinking as they develop an understanding of how to use the circumference to find the radius and diameter.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.6.1 Reasonableness

Prompt students to verify their answers to each question as means for self-regulation by asking themselves the following questions:

- Is my answer reasonable in the context?
- Does my answer make sense in this context?
- If not, why? What could have happened to make it unreasonable?

DIFFERENTIATE INSTRUCTION

In 4.7.3 Guided Practice, provide a visual entry point by prompting students to use a different color for each dimension (Circumference, radius, diameter) and continue using that color throughout the series of substitutions.

For example: $C = 2 \pi r$, $C = \pi d$.

Consider reading the questions aloud or using the online text reader to provide an auditory entry point.

4.7.1 WARM-UP: EVERGLADES NATIONAL PARK

TEACHER MOVES

Consider asking students if they have ever heard of these two native Florida species. Consider contacting 7th grade science teachers to feature some of the native Florida species found in Everglades National Park.

ENRICHMENT

How do you convert from centimeters to millimeters or vice versa? Why did Andrew use different conversion factors?

TEACHER MOVES

Prompt students to reference their responses to question 1 after they complete question 2. Note any student responses that change based on their findings.



4.7.1 Warm-Up: Everglades National Park

The Florida Everglades is home to two native species that are considered to be close to extinction. Cape Sable seaside sparrows and the Florida leafwing butterflies live only within Everglades National Park and the immediately adjacent lands.

Cape Sable seaside sparrows are small birds about 13 centimeters long.



The Florida leafwing is a medium-sized butterfly approximately 77 millimeters in length.

- Which species has the largest length?
Cape Sable seaside sparrow has the largest length.

Andrew multiplies the sparrow’s length by 0.4 to estimate the length of the sparrow in inches. To estimate the length of the butterfly in inches he multiplies the length by 0.04.

- Use Andrew’s method to determine the length of the sparrow and the length of the butterfly in inches.

Cape Sable Seaside Sparrow Length (inches)	Florida Leafwing Length (inches)
5.2in	3.08in



4.7.2 Exploration: Using Circumference to Find Other Measurements

1. A circular tank at a wastewater treatment facility in Davie, Florida has walkways built around the edge of the tank and across the center of the tank so that water management workers can check and treat the water. The walkway built around the top of the tank, shown in pink, is 37.68 meters (m) long.



Discuss with your partner how this information could be used to find the distance of the walkway that runs across the top of the water tank. Write a summary of your discussion.

In order to find the distance that runs across the center of the tank we could use the relationship between the circumference and the diameter. See this relationship and calculation below.

$$C = \pi d$$

or

$$d = \frac{C}{\pi} = \frac{37.68}{3.14} = 12 \text{ m}$$

4.7.2 EXPLORATION: USING CIRCUMFERENCE TO FIND OTHER MEASUREMENTS

ENRICHMENT

What part of a circle does the walkway represent? What information is given and what is unknown? Which circumference formula should we use to answer this question?

TEACHER MOVES

Think-Pair-Share

Prompt students to initially try to answer the questions independently. Then, encourage them to discuss their work and answers with a partner. Finally, ask students who used a variety of representations and/or strategies to express their findings with partners and the class.

2. Josephine said the formula for the circumference of a circle could be used to find the diameter. Her partial explanation and work are shown. Work with your partner to complete Josephine's work and explanation by filling in the boxes.

Josephine's Work	Explanation
$C = \pi d$	Begin with the formula for circumference of a circle in terms of diameter.
$\boxed{37.68} = 3.14 \cdot d$	Substitute 37.68 for the value of the circumference (C) and 3.14 for pi.
$\frac{\boxed{37.68}}{3.14} = \frac{3.14}{\boxed{3.14}} d$	Isolate the diameter (d) by dividing both sides of the equation by 3.14 .
$\boxed{12 \text{ m}} = d$	Calculate the diameter.

3. Some tanks at wastewater treatment centers have walkways built only to the center of the tank.

- a. What part of a circle does this represent?
Radius



- b. What would be the length of a walkway like this in Davie, Florida?
6 m

ENRICHMENT

If you know the circumference of the tank, how can we find the radius? Which circumference formula did you use to solve the problem? What is the relationship between the diameter and radius?

4.7.3 GUIDED PRACTICE: USING CIRCUMFERENCE TO FIND OTHER MEASUREMENTS

TEACHER MOVES

Think-Pair-Share

Prompt students to initially try answering the questions independently. Then, encourage them to discuss their work and answers with a partner. Finally, ask students who used a variety of representations and/or strategies to share with the whole class.

DIFFERENTIATE INSTRUCTION

Provide a visual entry point by prompting students to use a different color for each dimension (Circumference, radius, diameter) and continue using that color throughout the series of substitutions. For example: $C=2\pi r$, $C=\pi d$. Consider reading the questions aloud or using the online text reader to provide an auditory entry point.



4.7.3 Guided Practice: Using Circumference to Find Other Measurements

- Pizzaria and Little Italy are both offering large pizzas for \$12.99 each. Victor knows that Pizzaria's pizza has a circumference of 43.96 inches (in.) and Little Italy's pizza has a diameter of 13.25 in. Which pizza has the largest diameter?

$$C = d\pi = d = \frac{C}{\pi} = \frac{43.96}{3.14} = 14\text{in}$$

Pizzaria's pizza has the largest diameter.

- Fredrick is a landscape architect and is updating a circular garden. He plans to put a fountain in the center of the garden but does not know the garden's radius. If the garden has a circumference of 54.95 feet (ft), what is the radius?

$$C = 2\pi r = r = \frac{C}{2\pi} = \frac{54.95}{3.14 \cdot 2} = 8.75\text{ft}$$

- Mary recently baked both an apple and cherry pie. The apple pie has a diameter of $23\frac{1}{2}$ centimeters (cm), and the cherry pie has a circumference of 71.78 cm.

- What is the radius of the cherry pie?

$$r = \frac{C}{2\pi} = \frac{71.78}{6.28} = 11.43\text{cm}$$

Radius of the cherry pie is 11.43cm

- Which pie has the larger radius?

$$\text{Apple pie radius} = r = \frac{d}{2} = \frac{23.5}{2} = 11.75\text{cm}$$

The apple pie has a larger radius than cherry pie.

4.7.4 YOUR TURN!

ENRICHMENT

Could you determine the circumference of the inner circle with the information provided? Why or why not?

QUESTIONING

Does your school or community have any circles to use in substitution of this question? If so, encourage students to visit that circle or research dimensions online.

For example, if your school crest or mascot has a circle in the logo, consider prompting your students to find its dimensions. Alternatively, students can use devices to research local examples of circles in their communities.



4.7.4 Your Turn!

- Fort Lauderdale recently installed a new traffic roundabout. At its widest part, the roundabout has a circumference of 345.4 feet (ft). What is the radius of the traffic roundabout at its widest part?



$$r = \frac{C}{2\pi} = \frac{345.4}{2 \times 3.14} = 55\text{ft}$$

- The widest part of a super bouncer ball has a circumference of $\frac{17\pi}{20}$ decimeters (dm).

- What is the diameter of the super bouncer ball?

$$d = \frac{\frac{17\pi}{20}}{\pi} = \frac{17}{20} = 0.85\text{dm}$$

- If the widest part of a regular bouncy ball has a circumference of 0.49π dm, what is the diameter?

$$d = \frac{C}{\pi} = \frac{0.49\pi}{\pi} = 0.49\text{dm}$$

- What is the difference in diameter of the super bouncer and the regular bouncy ball?

$$\text{difference in diameter} = 0.85\text{dm} - 0.49\text{dm} = 0.36\text{dm}$$

- The High Roller in Las Vegas, Nevada is the world's largest Ferris wheel. The High Roller has a circumference of 158.5π m. At the highest point on the Ferris wheel, how far from the center is the rider?

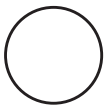
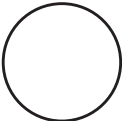


$$r = \frac{C}{2\pi} = \frac{158.5\pi}{2\pi} = 79.25\text{m}$$



4.7.5 Wrap-Up: Ticket Out the Door

1. Find the diameter and radius for circles with the given circumferences. Use $\pi = 3.14$. If necessary, round your answers to the thousandth.

Circle	 Circumference $\frac{5}{2}$ meters (m)	 Circumference 10.99 inches (in.)
Diameter	$d = \frac{c}{\pi} = \frac{2.5}{3.14} = 0.796m$	$d = \frac{c}{\pi} = \frac{10.99}{3.14} = 3.5in$
Radius	$r = \frac{d}{2} = \frac{0.7961783439}{2} = 0.398m$	$r = \frac{d}{2} = \frac{3.5}{2} = 1.75in$

4.7.5 WRAP-UP: TICKET OUT THE DOOR

QUESTIONING

Challenge students who finish early with the following question:

- Would it ever be possible for circumference to be a rational number? Provide justification (either written or algebraic) for your response.